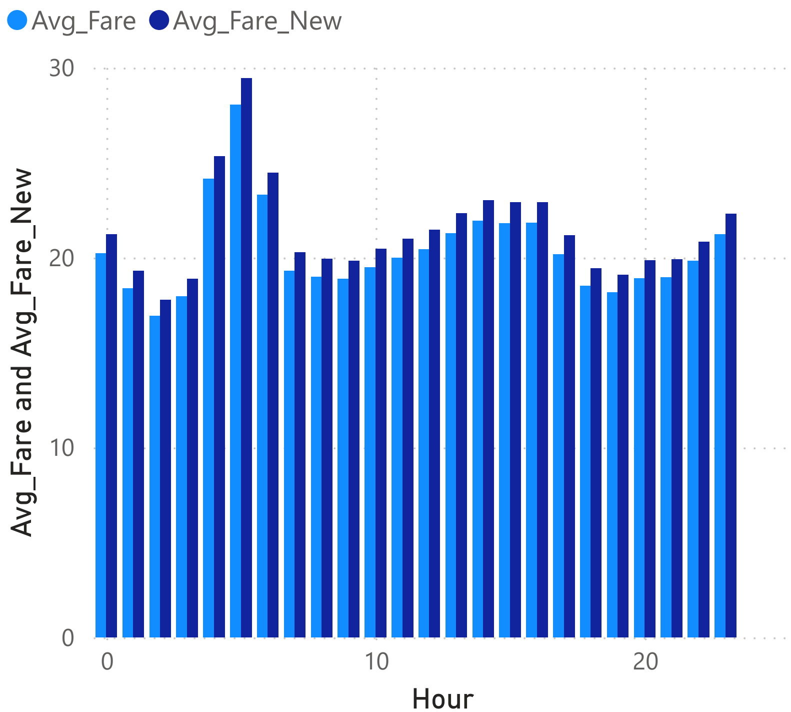
Q5) Pressure on the syste,

Distance vs Fare Behavior (Before vs After Change)



Average Fare (Before) Average Fare (After)

20.04	**21.05**
Avg_Fare	Avg_Fare_New

Parameter Chosen:
Fare amount was selected as it directly affects both passengers and drivers.

Intervention:
Fare was increased by R% to simulate a policy change.

Observation:
The average fare increased slightly, while the overall trip pattern remained stable.

System Response:
The system absorbs small changes without major disruption.
However, higher fare levels may reduce demand and amplify the effect.

Conclusion:
The system shows conditional stability — it absorbs small changes but may react strongly under larger variations.

The system absorbs small fare changes with minimal impact on trip patterns. However, as fares increase, trip demand may reduce, showing an amplified effect on the system. This suggests that trip behavior is sensitive to pricing changes.

Limitation:
This simulation assumes fare change does not immediately affect trip volume.
In reality, passenger behavior may change over time, leading to different system dynamics.